

MALACCA, Malaysia

WMO: 486650

Lat: 2.2551N

Lon: 102.2440E

Elev: 35

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	73.1	73.5	67.6	101.6	83.1	68.7	105.6	82.0	15.3	84.8	13.9	85.1	3.6	40	0.330

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.0	93.2	78.9	91.8	79.1	91.3	79.1	81.9	89.1	81.4	88.6	80.9	88.0	7.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	156.2	86.6	79.2	151.6	85.8	79.0	150.6	85.6	45.7	89.8	45.2	89.4	44.7	88.4	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	11.7	10.1	DB	70.7	95.6	1.6	2.1	69.6	97.1	68.6	98.3	67.7	99.5	66.5	101.0
			WB	68.5	83.5	1.9	1.5	67.1	84.6	66.0	85.4	64.9	86.3	63.5	87.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	81.6	82.8	83.4	83.5	83.3	82.8	81.9	81.8	82.0	82.4	81.9	81.4
	DBStd	1.80	1.93	1.89	1.88	1.70	1.56	1.56	1.58	1.53	1.45	1.49	1.39	1.76
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11822	978	918	1036	1004	1034	983	990	987	960	1003	956	973
	CDD65	6347	513	498	571	554	569	533	525	522	510	538	506	508
	CDH74	65996	4771	5022	6041	6066	6559	6025	5691	5550	5356	5553	4763	4599
CDH80	24899	1658	1995	2472	2459	2684	2379	2119	2018	1966	2071	1604	1474	
Wind	WSAvg	4.1	5.9	6.2	4.8	3.7	3.4	3.3	3.4	3.6	3.4	3.6	3.6	4.7
Precipitation	PrecAvg	76.6	3.5	3.4	5.4	7.1	6.7	6.2	7.2	7.7	6.8	7.9	9.1	6.2
	PrecMax	103.1	10.7	11.0	14.2	15.4	16.8	18.3	18.6	19.3	14.4	13.7	16.7	19.4
	PrecMin	46.0	0.2	0.0	0.3	0.5	1.0	0.0	2.0	2.5	0.8	2.3	3.9	0.6
	PrecStd	10.6	2.6	2.7	2.9	3.1	3.1	3.7	3.3	3.6	2.8	3.1	3.0	3.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	94.5	95.0	94.6	93.4	91.8	91.4	91.3	91.5	91.8	91.5	91.4
		MCWB	77.9	77.1	77.8	80.4	80.5	79.7	79.3	79.5	79.0	78.8	79.3	78.9
	2%	DB	90.6	92.6	93.0	92.2	91.7	91.0	89.9	89.8	90.0	90.8	89.9	89.7
		MCWB	77.6	77.5	78.4	80.1	80.3	79.6	79.1	79.1	79.0	78.9	79.2	78.6
	5%	DB	89.3	91.2	91.4	91.1	90.9	89.8	89.2	89.0	89.3	89.7	88.9	88.4
		MCWB	77.6	77.6	78.9	80.0	80.1	79.5	79.0	79.0	78.9	78.9	79.1	78.4
	10%	DB	87.8	89.5	89.8	89.6	89.6	89.0	88.0	87.9	88.0	88.2	87.6	87.3
		MCWB	77.4	77.6	78.8	79.8	79.9	79.5	78.9	79.0	78.9	78.8	79.1	78.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.4	81.3	81.8	82.6	82.6	82.0	81.6	82.1	81.5	81.5	81.6	81.3
		MCDB	88.5	89.6	89.9	90.6	90.1	88.4	88.0	88.0	88.4	89.1	88.6	88.7
	2%	WB	79.9	80.0	80.8	81.6	81.7	81.2	80.7	80.7	80.5	80.4	80.6	80.1
		MCDB	87.1	88.1	88.8	89.4	88.8	87.8	87.2	87.2	87.4	87.6	87.6	86.9
	5%	WB	79.0	79.2	80.1	80.9	81.0	80.6	80.0	79.9	79.9	79.8	79.9	79.3
		MCDB	86.0	87.3	87.9	88.4	88.0	87.2	86.4	86.3	86.5	86.8	86.6	85.9
	10%	WB	78.1	78.5	79.5	80.1	80.5	80.0	79.5	79.3	79.3	79.2	79.2	78.7
		MCDB	84.9	86.5	87.1	87.3	87.3	86.4	85.7	85.5	85.7	86.0	85.6	85.0
Mean Daily Temperature Range	5% DB	MDBR	12.9	14.6	14.5	13.7	13.0	12.6	12.5	12.3	12.8	13.1	12.8	12.1
		MCDBR	14.9	16.5	16.0	15.2	14.5	13.6	13.7	13.6	14.1	14.3	14.2	13.9
		MCWBR	5.3	5.5	5.6	5.7	5.5	5.2	5.2	5.6	5.5	5.4	5.7	5.4
	5% WB	MCDBR	13.4	14.8	14.6	14.0	13.4	12.8	12.7	12.6	13.2	13.5	13.3	12.9
	MCWBR	5.8	6.0	5.9	5.9	5.7	5.6	5.6	5.8	5.8	5.7	5.9	5.6	
Clear-Sky Solar Irradiance	taub	0.470	0.511	0.528	0.505	0.488	0.522	0.503	0.541	0.562	0.587	0.479	0.455	
	taud	2.277	2.134	2.099	2.174	2.224	2.122	2.183	2.063	2.004	1.924	2.254	2.332	
	Ebn at Noon	272	263	258	258	255	243	249	246	246	240	268	275	
	Edh at Noon	44	52	54	49	45	49	46	54	58	63	45	41	
All-Sky Solar Radiation	RadAvg	1461	1672	1675	1640	1551	1489	1474	1502	1489	1491	1428	1350	
	RadStd	126	131	117	69	63	69	73	65	106	111	78	89	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (4 neighbors)	+0.61	N/A	N/A	N/A	+0.54	+0.54	N/A	N/A	N/A	N/A
	+0.72	+0.32	N/A	+0.79	+0.92	+0.79	N/A	N/A	+275	+275

Nomenclature: See separate page